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Section 3.4 BSIT

Course: IT Elective 2
Date: 11-13-26

Activity 3: Student Registration Form

Code:

Program.cs

StudentRegistration

StudentRegistratio

```
using System;
using System.Collections.Generic;
using System.Drawing;
using System.Windows.Forms;

namespace StudentRegistration
{
    // ═══════════════════════
    // Part 2: Classes (OOP)
    // ═══════════════════════
    // Base class that represents any person.
    public class Person
    {
        public string Name { get; set; }

        public Person(string name)
        {
            Name = name;
        }

        // Virtual so derived classes (like Student) can override it.
        public virtual string GetSummary()
        {
            return "Person: " + Name;
        }
    }

    // Derived class that represents a student.
    public class Student : Person
    {
        // Encapsulation: private setter
        public string StudentID { get; private set; }

        public string Course { get; set; }

        // Passes the name to the base class constructor with : base(name).
        public Student(string studentId, string name, string course)
            : base(name)
        {
            StudentID = studentId;
            Course = course;
        }

        // Override the base summary.
        public override string GetSummary()
        {
            return "Saved! Student ID: " + StudentID +
                   " | Name: " + Name +
                   " | Course: " + Course;
        }
    }
```

Program.cs

StudentRegistration

```
53      // ═══════════════════
54      // Part 1: Form Design
55      // ═══════════════════
56      public class MainForm : Form
57      {
58          // Controls (the (Name) of each control is set exactly as required)
59          private GroupBox studentGroupBox;
60          private Label studentIdLabel;
61          private TextBox studentIdTextBox;
62          private Label nameLabel;
63          private TextBox nameTextBox;
64          private Label courseLabel;
65          private ComboBox courseComboBox;
66          private Button saveButton;
67          private Button clearButton;
68          private Label feedbackLabel;
69
70          // In-memory storage for the students.
71          private readonly List<Student> students = new List<Student>();
72
73          public MainForm()
74          {
75              InitializeFormControls();
76          }
77
78          // Form layout: consistent alignment and spacing.
79          private void InitializeFormControls()
80          {
81              // Form settings
82              Text = "Student Registration";
83              Font = new Font("Cascadia Code", 10F);
84              ClientSize = new Size(440, 330);
85              FormBorderStyle = FormBorderStyle.FixedDialog;
86              MaximizeBox = false;
87              StartPosition = FormStartPosition.CenterScreen;
88
89              // Group box
90              studentGroupBox = new GroupBox
91              {
92                  Name = "studentGroupBox",
93                  Text = "Student Information",
94                  Location = new Point(15, 15),
95                  Size = new Size(410, 170)
96              };
97
```

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StudentRegistration

StudentRegistration.Program

```
97
98          // Row 1: Student ID
99          studentIdLabel = new Label
100         {
101             Name = "studentIdLabel",
102             Text = "Student ID:",
103             Location = new Point(15, 37),
104             AutoSize = true
105         };
106         studentIdTextBox = new TextBox
107         {
108             Name = "studentIdTextBox",
109             Location = new Point(130, 34),
110             Size = new Size(260, 27)
111         };
112
113         // Row 2: Student Name
114         nameLabel = new Label
115         {
116             Name = "nameLabel",
117             Text = "Student Name:",
118             Location = new Point(15, 82),
119             AutoSize = true
120         };
121         nameTextBox = new TextBox
122         {
123             Name = "nameTextBox",
124             Location = new Point(130, 79),
125             Size = new Size(260, 27)
126         };
127
128         // Row 3: Course
129         courseLabel = new Label
130         {
131             Name = "courseLabel",
132             Text = "Course:",
133             Location = new Point(15, 127),
134             AutoSize = true
135         };
136         courseComboBox = new ComboBox
137         {
138             Name = "courseComboBox",
139             Location = new Point(130, 124),
140             Size = new Size(260, 27),
141             DropDownStyle = ComboBoxStyle.DropDownList
142         };
143         courseComboBox.Items.AddRange(new object[] { "BSIT", "BSCS", "BSIS" });
144
145         studentGroupBox.Controls.AddRange(new Control[]
146         {
147             studentIdLabel, studentIdTextBox,
148             nameLabel, nameTextBox,
149             courseLabel, courseComboBox
150         });
151
```

Program.cs

StudentRegistration

```
152             // Buttons
153             saveButton = new Button
154             {
155                 Name = "saveButton",
156                 Text = "Save",
157                 Location = new Point(225, 200),
158                 Size = new Size(95, 34)
159             };
160             clearButton = new Button
161             {
162                 Name = "clearButton",
163                 Text = "Clear",
164                 Location = new Point(330, 200),
165                 Size = new Size(95, 34)
166             };
167
168             // Feedback label
169             feedbackLabel = new Label
170             {
171                 Name = "feedbackLabel",
172                 Text = "",
173                 Location = new Point(15, 250),
174                 Size = new Size(410, 65),
175                 AutoSize = false
176             };
177
178             Controls.AddRange(new Control[]
179             {
180                 studentGroupBox, saveButton, clearButton, feedbackLabel
181             });
182
183             // Pressing Enter triggers Save.
184             AcceptButton = saveButton;
185
186             // Part 3: connect the event handlers to the Click events
187             saveButton.Click += saveButton_Click;
188             clearButton.Click += clearButton_Click;
189         }
190
```

Program.cs

StudentRegistration

StudentRegistration.Progra

```
191         // ═══════════════════════════════════════════
192         // Part 3 + Part 4: Event handlers and input validation
193         // ═══════════════════════════════════════════
194
195         1 reference
            private void saveButton_Click(object sender, EventArgs e)
196         {
197             // Read and trim the text from the input controls.
198             string studentId = studentIdTextBox.Text.Trim();
199             string studentName = nameTextBox.Text.Trim();
200
201             // Validation: Student ID must not be empty or only spaces.
202             if (string.IsNullOrWhiteSpace(studentId))
203             {
204                 ShowError("Please enter a student ID.");
205                 studentIdTextBox.Focus();
206                 return;
207             }
208
209             // Validation: Student Name must not be empty or only spaces.
210             if (string.IsNullOrWhiteSpace(studentName))
211             {
212                 ShowError("Please enter a student name.");
213                 nameTextBox.Focus();
214                 return;
215             }
216
217             // Validation: a course must be selected (-1 means nothing selected).
218             if (courseComboBox.SelectedIndex == -1)
219             {
220                 ShowError("Please select a course.");
221                 courseComboBox.Focus();
222                 return;
223             }
224
225             string course = courseComboBox.Text;
226
227             // Create the Student object and keep it in memory only.
228             Student student = new Student(studentId, studentName, course);
229             students.Add(student);
230
231             // Show the summary as feedback to the user.
232             feedbackLabel.ForeColor = Color.DarkGreen;
233             feedbackLabel.Text = student.GetSummary();
234         }
235
```

Program.cs*

StudentRegistration

StudentRegistration.Stu

```
1 reference
        private void clearButton_Click(object sender, EventArgs e)
        {
            // Empty the text boxes, reset the course selection, clear feedback.
            studentIdTextBox.Clear();
            nameTextBox.Clear();
            courseComboBox.SelectedIndex = -1;
            feedbackLabel.Text = "";

            // Put the cursor back on the first field for the next entry.
            studentIdTextBox.Focus();
        }

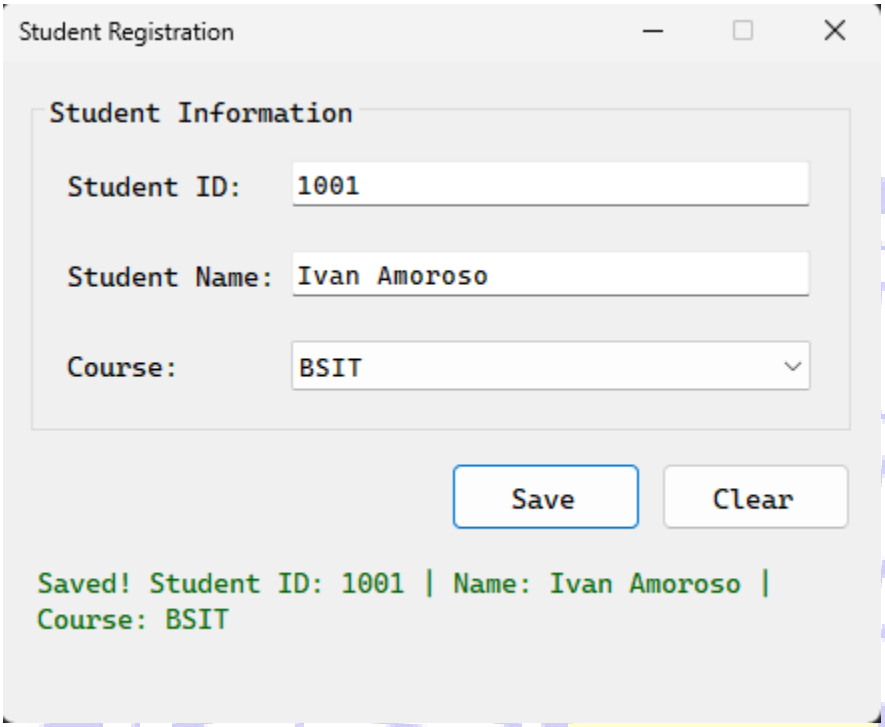
        // Displays an error message in red so it is easy to notice.
        3 references
        private void ShowError(string message)
        {
            feedbackLabel.ForeColor = Color.Firebrick;
            feedbackLabel.Text = message;
        }
    }

    // ═══════════
    // Entry point
    // ═══════════

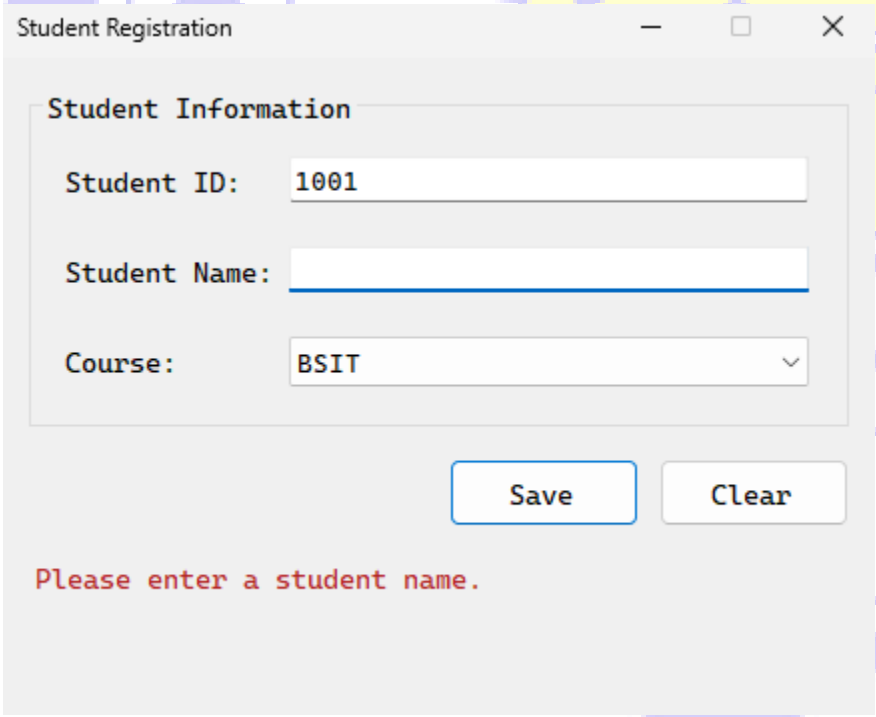
    0 references
    internal static class Program
    {
        [STAThread]
        0 references
        static void Main()
        {
            Application.EnableVisualStyles();
            Application.SetCompatibleTextRenderingDefault(false);
            Application.Run(new MainForm());
        }
    }
}
```

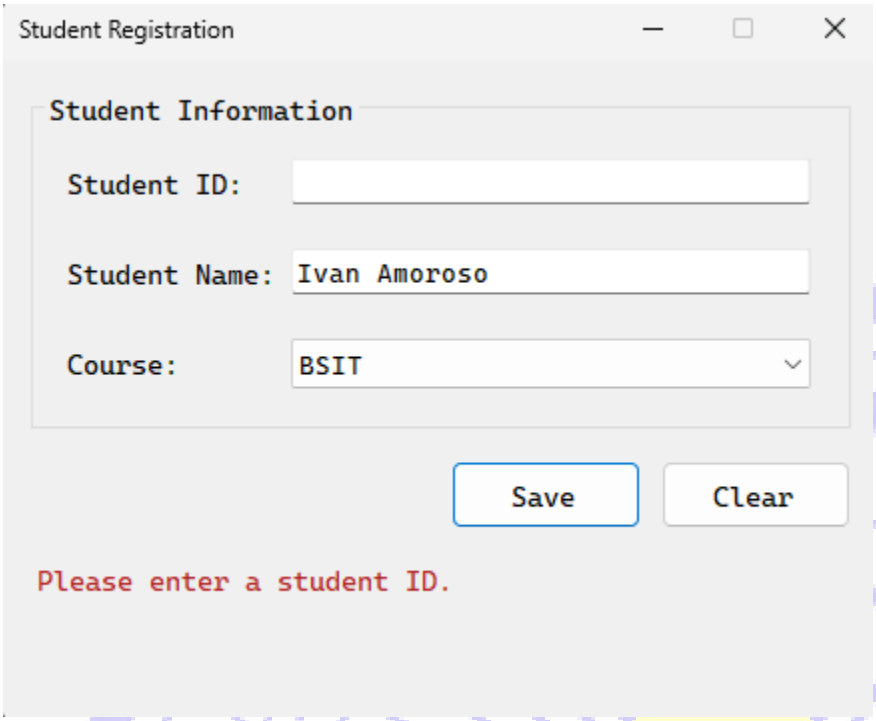
Output:

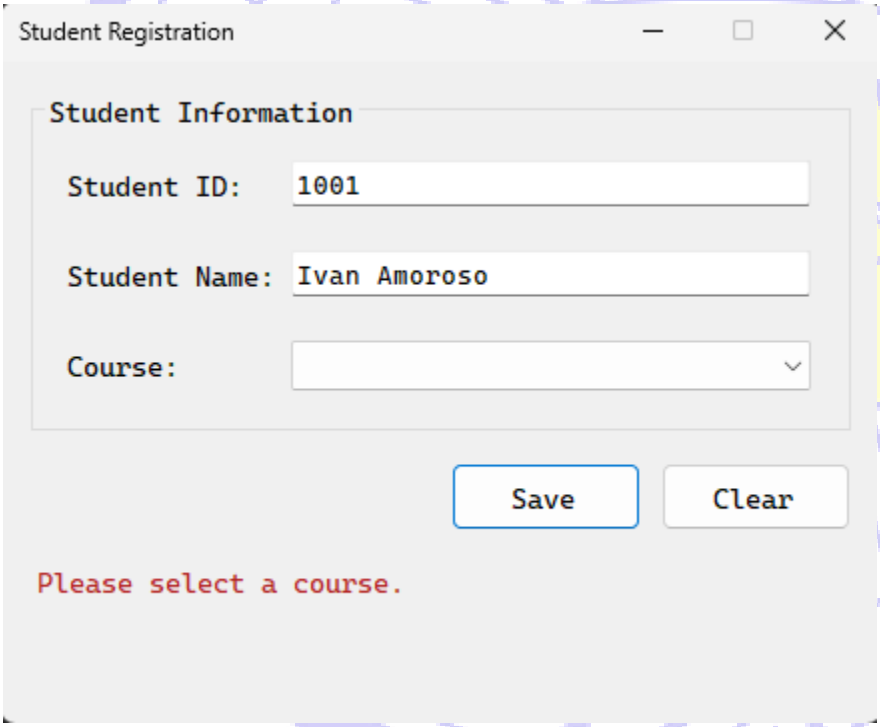
A: Successful



B: Error




Student Registration
Student Information
Student ID:
Student Name: Ivan Amoroso
Course: BSIT
Save
Clear
Please enter a student ID.


Student Registration
Student Information
Student ID: 1001
Student Name: Ivan Amoroso
Course:
Save
Clear
Please select a course.